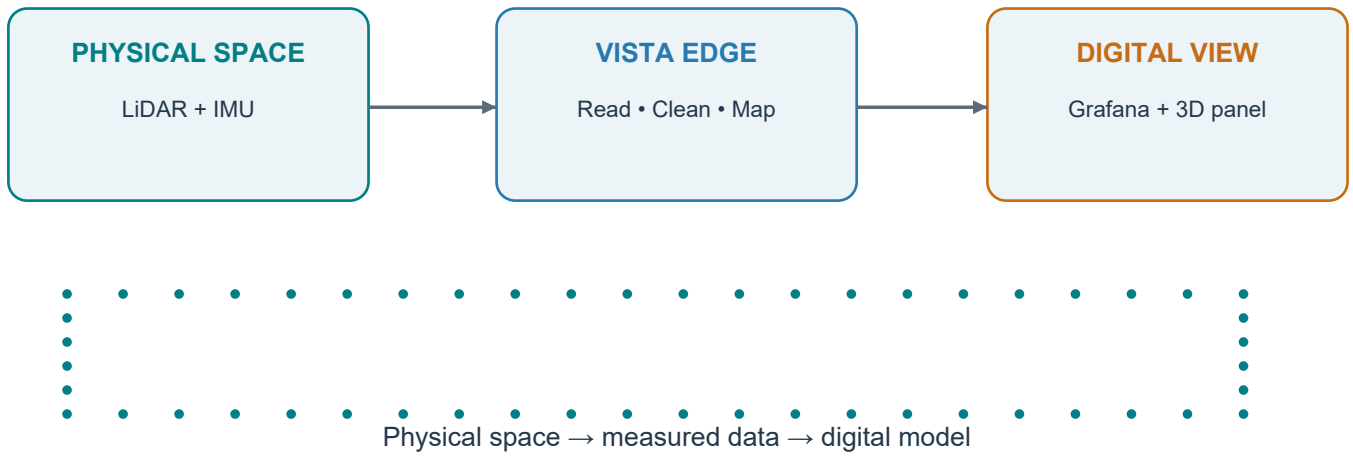


01 / VISTA EDGE

FROM LIDAR TO DIGITAL TWIN

Project Guide / Architecture, Development and Operations



VISTA converts LiDAR and IMU measurements into a live 3D view, an accumulated room map, and operational telemetry. Its long-term goal is a digital twin: a persistent representation of a real space, later enriched with moving objects, tracking and activity analytics.

Audience and scope

This English guide is written for newcomers, project members and deployment engineers. It distinguishes what the code implements today from research options and capabilities that still need real-device validation.

Edition	Evidence
Updated	6 October 2026 - Australia/Sydney
Implementation	Current C++ source, public DeviceConfig, V4 dashboard, plugin manifests and recorded verification results
Source baseline	Repository HEAD f9e9a01 plus the inspected working-tree changes; HEAD alone does not reproduce this snapshot
Artifact	Standalone document outside the code repository. No credentials or secret-file contents are included.

02 / Reading map

Start with the overview; use the technical sections when you configure or extend the system.

Sections / pages	What you will learn
3-7	Purpose, current capabilities, four-layer architecture, data flow and folders
8-11	Workers, Pub-Sub, topic contracts and device acquisition
12-17	Coordinates, preprocessing, host-clock alignment, LIO estimation, states and recovery
18-23	Ground validation, Room Map lifecycle, free-space evidence, tiles/LOD and saving
24-28	Grafana data paths, installation, dashboard behavior, Ethernet deployment and first-run checks
29-34	Configuration reference: sensor, ground, LIO, map, cache, streaming and safe tuning
35-38	Windows, modern Linux, legacy ADLINK and verification evidence
39-42	Troubleshooting, operations, roadmap and glossary
43-46	Repository evidence and clickable research/official references

Recommended reading paths

- **New team member:** sections 3-7, 20-24 and 42. Learn what a scan, pose, map and dashboard actually represent.
- **Mapping developer:** sections 9-23 and 29-34. Follow measurement timing and coordinate contracts before changing algorithms.
- **Deployment engineer:** sections 25-28 and 35-40. Build on the right target and test both network paths independently.

Status convention

IMPLEMENTED: present in inspected code. **PLANNED:** not yet a running feature. **NEEDS VALIDATION:** software checks exist but the relevant hardware, motion or endurance trial is not established.

03 / The project in five minutes

A point cloud is a set of measured 3D positions, not a camera image or a complete room model.

A LiDAR repeatedly samples visible surfaces. Each scan captures only what the sensor sees at that time. VISTA has two complementary outputs: a responsive live sensor view and a persistent room map that combines repeated observations. When the sensor moves, local LiDAR-inertial odometry estimates the pose needed to place those measurements in the world.

Output	Purpose	What it does not prove
Live sensor cloud	Inspect current measurements, including the floor, without waiting for localization.	A valid world position or a complete room.
Localization pose	Estimate device position/orientation using IMU prediction and LiDAR correction.	Absolute ground-truth accuracy, loop closure or globally consistent SLAM.
Room Map	Accumulate confirmed geometry; keep floor, walls and furniture; store large maps on disk.	Semantic recognition or automatic removal of every person.
Ground-processed cloud	Separate floor-related points for future object-oriented applications.	Implemented detection or tracking.
Grafana telemetry	Observe input, mapping, localization and worker/resource health.	A durable time-series database or a safety-certified monitoring system.

Two operating modes

The architecture supports both stationary mapping with LioEnabled=0 and moving mapping with LioEnabled=1. Movement requires correct timing, IMU units, extrinsics and observable geometry, not merely a changing IMU quaternion.

04 / Current capabilities and boundaries

Treat this as a development system with explicit evidence, not an already validated commercial product.

Status	Capability
IMPLEMENTED	C++17 Windows/Linux program; Quanergy M8, RealSense L515 and Unitree L2 driver paths; typed Pub-Sub and worker monitoring.
IMPLEMENTED	Independent live cloud; Unitree IMU publication; ROS-free FAST-LIO adaptation; host-monotonic alignment; stationary initialization and local pose correction.
IMPLEMENTED	Static/RANSAC/hybrid ground handling; disk-backed Room Map; repeated-hit confirmation; conservative free-space clearing; camera-dependent LOD; full confirmed PCD export.
IMPLEMENTED	Grafana numeric telemetry, lidarpointcloud 3D panel, VISTA Live data source with heartbeat expiry and reconnection.
PLANNED	Object clustering/detection/tracking, velocity and occupancy analytics, camera/radar fusion, semantic digital-twin layers.
NEEDS VALIDATION	Real Unitree movement, ADLINK USB/serial performance, absolute map accuracy, loaded-map localization, power-loss recovery and long-duration operation.

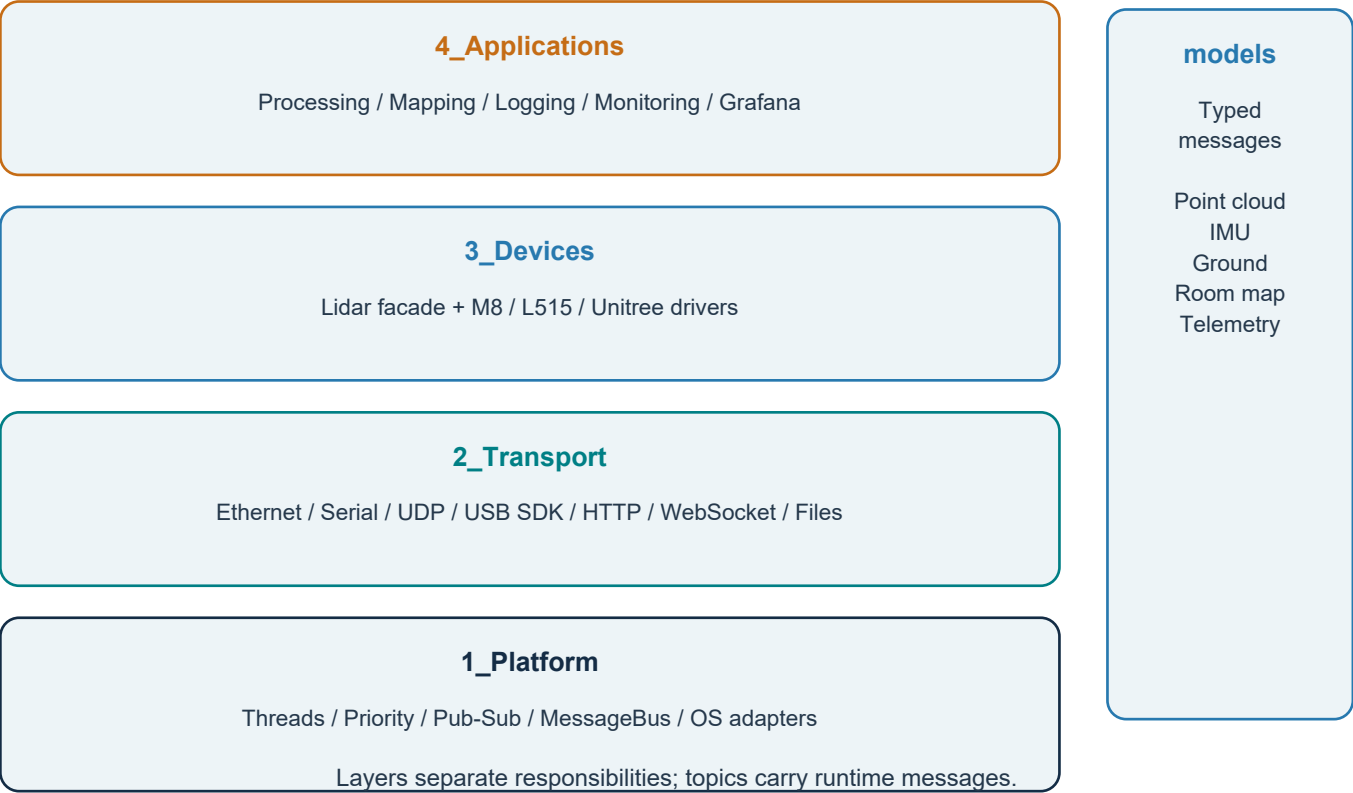
Important distinctions

- Local odometry is implemented. Global loop closure and automatic relocalization into a loaded PCD are not.
- The upstream FAST-LIO code is adapted into a private numerical target, not launched as its ROS node.
- Rust under embedded is a separate implementation and does not have demonstrated parity with the current C++ LIO/Room Map pipeline.

Algorithm background: FAST-LIO2 and ikd-Tree [R01-R02]. VISTA-specific behavior is established by the repository, not the papers.

05 / The four-layer architecture

The original colored architecture illustration is retained below.

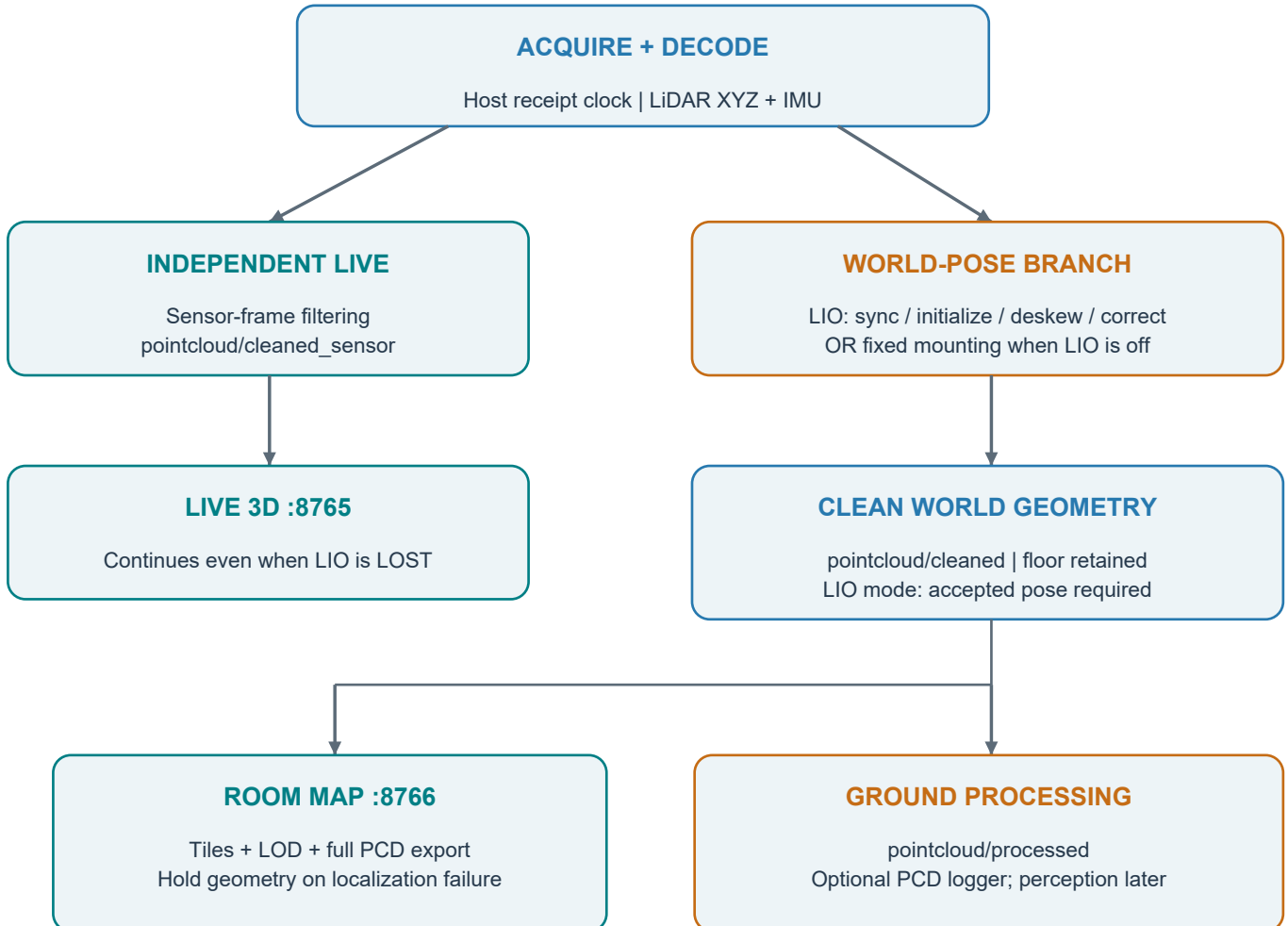


Layer	Responsibility
1_Platform / navy	OS adaptation, worker lifecycle, threading/priority, bounded Pub-Sub, MessageBus and WaitSets.
2_Transport / teal	Serial, TCP/UDP, SDK USB, HTTP, WebSocket and storage I/O.
3_Devices / blue	LiDAR abstraction, driver selection, protocol validation/decoding and measurement publication.
4_Applications / orange	Live/world preprocessing, LIO, ground, Room Map, logging, monitoring and Grafana output.

models provides shared typed contracts; it is not a fifth execution layer. Runtime messages cross topic boundaries; applications should not absorb low-level serial/socket APIs.

06 / The current end-to-end flow

Live visibility and world-map correctness are deliberately separated.



Failure isolation

In LIO mode, invalid localization blocks new world-map geometry, not the independent live view. Existing map points are kept. There is no silent fixed-pose fallback that could smear moving scans into the room.

The decoder also publishes lidar/imu. Optional RAW logging branches from lidar/raw. Optional processed PCD logging branches after ground processing; it is not the deduplicated Room Map export.

07 / Repository navigation

All paths in this guide are relative to D:/MASTER/VISTA/LiDar/Repo unless a full host path is shown.

Location	What to inspect or change
embedded_C/main.cpp + lib.hpp	Explicit wiring and worker startup; main includes lib.hpp.
embedded_C/config.cpp/.hpp	TXT parser, validation, runtime filters/queues and worker policies.
embedded_C/DeviceConfig.txt	Public run settings. The executable reads DeviceConfig.txt from its working directory.
embedded_C/1_Platform	Concurrency, message bus, OS helpers and measurement clock.
embedded_C/2_Transport	Peripheral interfaces, messaging and storage, including tiles/PCD.
embedded_C/3_Devices/lidars	Common LiDAR interfaces and M8/L515/Unitree implementations.
embedded_C/4_Applications/mapping	lio and room_map; pose estimation is distinct from map accumulation.
embedded_C/4_Applications	processing, logger, monitoring, grafana_bridge and pointcloud_websocket.
embedded_C/models + unittest	Independent topic models and software tests.
dashboard	VISTA_Live_Dashboard_V4_Free_Space.json; plugins/lidarpointcloud and plugins/vistalive.
third_party	Pinned shared SDK/dependency submodules. Generated adaptations stay in build directories.
data/raw, data/processed, data/maps	Optional capture files, processed scans, room-map PCDs and spatial tile directories.
embedded	Separate Rust crate, not the current C++ deployment executable.

08 / Workers, priority and lifecycle

One worker is currently one `std::thread`, not a thread-pool task.

Worker	Level	Function / activation
lidar-read	1	Connect/read/reconnect; publish raw receipt.
lidar-decode	2	Protocol validation; decoded scans and IMU.
raw-logger / pcd-logger	2	Optional recording workers, independently enabled.
live-pointcloud-preprocessing	3	Sensor-frame live filtering, always created.
pointcloud-preprocessing	3	Clean world branch; apply mount only in fixed mode.
ground-processing	3	Ground calibration, removal and diagnostics.
lidar-inertial-odometry	3	Created when <code>LioEnabled=1</code> .
system-monitor / room-mapping	4	Resource health; map build/load/disabled/error status.
grafana-bridge / pointcloud-websocket	4	Created when <code>GrafanaEnabled=true</code> .
room-map-websocket	5	Camera-dependent map view, gated by <code>GrafanaEnabled</code> .

With Grafana enabled and both loggers disabled, there are 10 workers in fixed mode or 11 with LIO. RAW logging adds one and processed PCD logging adds another. The inspected configuration has LIO and RAW logging on: normally 12 workers.

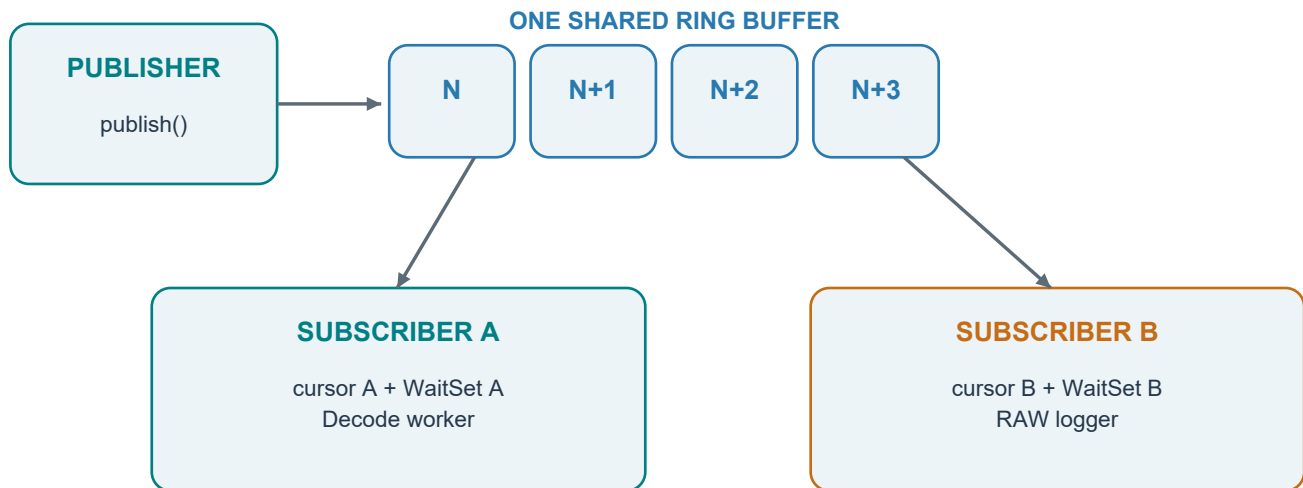
Levels 1-5 map to relative OS priorities: Windows HIGHEST through LOWEST; Linux `SCHED_OTHER` nice values 0, 2, 5, 10 and 15. Smaller level numbers mean higher priority, not hard real-time execution.

Lifecycle

Ctrl+C or SIGTERM requests cooperative stop, closes/wakes topics and joins workers. Stop Debugging, forced termination or power loss may bypass final map export. There is no separate Room Map logging worker.

09 / Pub-Sub, queues and independent readers

A topic is a broadcast channel; consumers do not remove each other's messages.



Subscribers retain immutable messages; reading does not remove data for other readers.

- Each topic uses a bounded ring. When full, the oldest slot is replaced; the publisher does not wait for every consumer.
- Each subscriber owns a cursor and drop count. A slow subscriber skips overwritten messages; faster subscribers continue independently.
- Consumers retain `shared_ptr<const Message>`. Reading does not copy the entire point vector or make it mutable. A transformation creates its own result.
- WaitSets let a worker wait for multiple input topics without spinning. Shutdown wakes waiting consumers.

Backpressure trade-off

Finite queues bound backlog, not processing time or total process RAM. A retained message remains allocated after its ring slot changes. Drops must be interpreted per consumer; a smooth live panel does not prove lossless RAW logging or LIO.

10 / Topic and model contracts

Data types distinguish sensor observations, estimated world scans and persistent geometry.

Topic	Role / contract
lidar/raw	Transport payload plus receipt timing; decoder and optional RAW logger.
pointcloud/decoded	Measured sensor cloud with per-point timing/metadata; input to live filtering and LIO.
lidar/imu	Independent ImuMessage from models/imu/imu_message.hpp. One active IMU producer per session.
pointcloud/cleaned_sensor	Finite/range/voxel-filtered sensor XYZ; floor kept, no world ROI/mount/ground removal.
localization/pointcloud_world	Accepted motion-compensated world observations; never guessed LOST poses.
localization/status	Independent LocalizationStatus: state, pose validity, quality and Reason.
pointcloud/cleaned	World geometry before ground removal; map and ground consumers.
pointcloud/processed	World cloud after ground processing; optional PCD logger.
processing/ground_status	Independent GroundStatusMessage under models/mapping.
mapping/room_map	Independent RoomMapMessage, bounded overview; not the entire map.
mapping/room_map_view	Source handle/revision for camera-dependent disk-backed LOD.
mapping/room_map_status	Lifecycle, geometry, cache, timing and I/O diagnostics.
monitor/*	System/worker/storage/power telemetry models; dashboard consumes selected fields.

IMU, ground, localization and Room Map models are independent from LiDAR envelopes. Persistent RoomMapPoint owns XYZ/intensity; acquisition-time ray origins remain on measured scans.

11 / LiDAR and IMU acquisition

Driver interfaces hide transport differences from applications.

Device	Connection / measurements
Unitree L2	Serial or UDP path; decoded cloud and integrated IMU share one timing adapter. The application does not require the vendor Linux binary or ROS node.
Quanergy M8	TCP Ethernet driver path; verify configured sensor endpoint and protocol framing.
RealSense L515	USB depth frames through the pinned librealsense SDK, converted to XYZ. This optional backend is disabled in the camera-free legacy ADLINK build.

Unitree serial checklist

- Windows uses a COM name; Linux uses an actual `/dev/ttyACM*` or `/dev/ttyUSB*` path. Verify the device and access permissions.
- `BaudRate(unitree-l2)=4000000`. A USB/hub/adaptor must support the actual path; the j5create model has not been validated in this guide.
- Only one program can normally own a serial port. Do not run the vendor viewer, replay tool and `vista_edge` against the same port simultaneously.
- IMU publishes quaternion, angular velocity in rad/s and acceleration including gravity in m/s^2 . Orientation is not a direct position measurement.

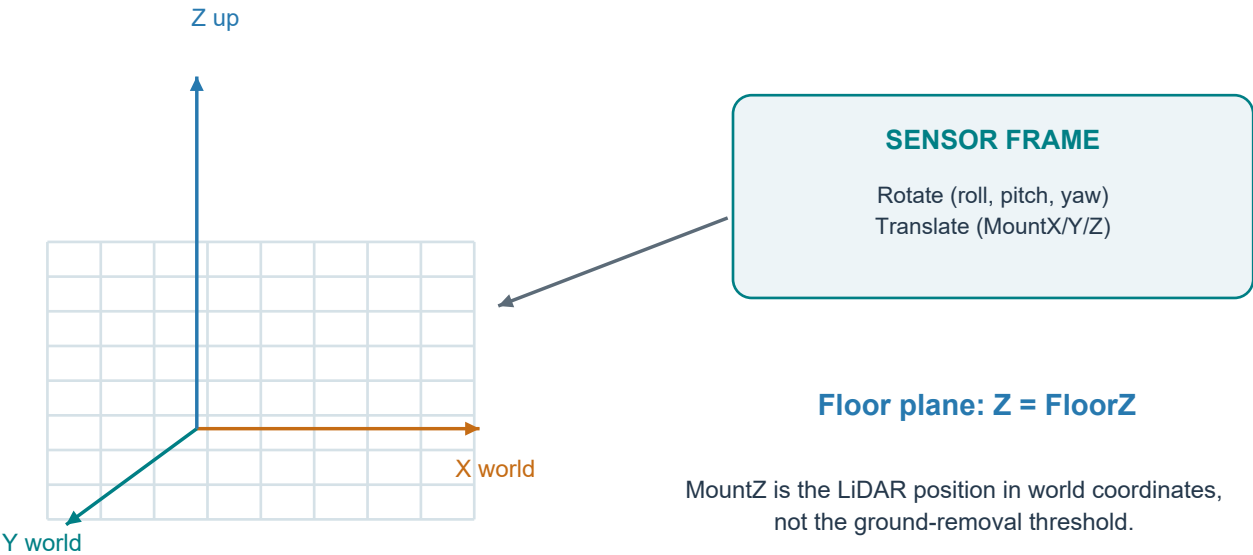
Configuration snapshot versus hardware

The inspected source TXT currently says `SerialPort(unitree-l2): COM6`, while earlier hardware trials used COM4. Set the actual port in the configuration used by your running executable; do not assume the source and build-directory copies are identical.

Protocol/API reference: Unitree SDK2 [R03]. Pinned optional SDK dependency: repository third_party/README.md.

12 / Coordinates, mounting and extrinsics

A coordinate mistake can look like a filtering, localization or rendering failure.



Transform	Meaning
MountX/Y/Z + roll/pitch/yaw	Fixed mode: continuing LiDAR-to-world transform. LIO mode: initial LiDAR pose in world only.
LiDAR-to-IMU XYZ/RPY	Fixed physical pose of the LiDAR in the IMU frame. Separate from mounting/world origin.
Estimated pose	Time-varying LIO state used to deskew/place accepted measurements in world.
FloorZ	World-frame floor prior. It is not LiDAR height or a sensor-frame Z threshold.

World preprocessing must not apply the mount transform a second time to LIO world scans. Free-space rays require the sensor origin at acquisition time; moving scans can carry a different world origin per point.

Measure rather than infer

On a table or the floor, measure the optical-center position. MountZ is not automatically zero because the housing touches the floor. A roll of 180 degrees means a flipped orientation only if that matches the physical axes.

13 / Preprocessing: what is filtered?

Three voxel sizes exist because three different tasks have different accuracy and load requirements.

Stage	Current behavior
Independent live branch	Finite coordinates, sensor range and 5 cm voxel filtering. No mounting, world ROI or ground removal.
World branch	Finite/range filtering, fixed mounting only when LIO is off, optional world ROI and 5 cm voxel filtering.
LIO scan preparation	Its own scan voxel and max-point budget for matching; starts from decoded scans, not ground-removed clouds.
Persistent Room Map	5 cm cells with repeated-window evidence; independent of LIO matching density.

Current `runtime_config()` sets preprocessing range to 0.1-200 m, ROI unset and voxel size to 0.05 m. These are currently code-level runtime defaults, not separate public TXT keys. They are software limits, not claims about the LiDAR's physical range.

World cleaning retains measured endpoints and ray metadata rather than synthesizing centroid rays. The Room Map can store a centroid as its representative without pretending that the centroid was a measured beam endpoint.

Why the live view may seem sparse

- A live panel shows one current filtered scan, not all historical points. A non-repeating scan pattern can fill a room only over time.
- Sensor occlusion, low reflectivity, valid-range rejection, voxel filtering and rendering budgets each reduce visible points.
- The 60 x 60 m grid is a visual reference, not a clipping boundary. Change camera zoom and distinguish live versus accumulated map.

Optional future SOR/ROR noise filters should be evaluated separately [R15]; they are not claimed as active here.

14 / The common host-clock model

Synchronization follows the Windows/ADLINK monotonic timeline, not calendar time.

Time field	Use
received_monotonic_ns	Host steady-clock stamp captured at transport receipt, before queue/decode delay.
measurement_timestamp_ns	MeasurementClock maps device-relative timing onto the host timeline. Used for alignment/propagation.
Original sensor timestamp	Relative intervals and reset/order validation; not a universal cross-device absolute epoch.
Host calendar timestamp	Grafana display and log/session filenames only. NTP/manual date changes must not alter LIO propagation.

Unitree clouds and its integrated IMU share one adapter. Sensor deltas preserve real spacing even when USB batches many samples into one read. Bounded drift correction follows the host without compressing acquisition intervals to decode-time intervals.

A future external IMU must stamp receipt in its own transport, align its own sensor deltas and publish the same ImuMessage to lidar/imu. Run only one IMU producer. A source-ID change is rejected rather than mixed into the current filter.

Timing quality matters

A common host clock is not hardware synchronization. Independent USB paths can have different latency. Receipt-only timing cannot reconstruct hidden batch acquisition times; external sensors still need timing/calibration validation.

LiolmuTopic, LiolmuSource, LioClockMode and LiolmuTimeOffsetMilliseconds are no longer public config switches. The current pipeline owns this clock contract.

15 / How LiDAR-inertial odometry works

IMU predicts motion; LiDAR corrects the prediction against observed surfaces.

Step	What happens
1. Stationary initialization	Require valid stable IMU samples and at least 0.5 s. Estimate initial gravity/bias state and a fixed acceleration scale.
2. Temporal coverage	Require IMU coverage for scan start/end and the actual integrated propagation horizon. Reject missing/gapped/nonmonotonic input.
3. IMU prediction	Propagate orientation, velocity, position, biases and covariance. Raw integration alone would drift.
4. Deskew	Use per-point acquisition times to compensate motion within the scan.
5. LiDAR correction	Find neighboring local-map surfaces; fit PCA planes; update through the iterated error-state filter.
6. Quality gate	Check effective matches, residual, geometry/covariance, finite state and pose steps.
7. Accepted publication	Publish world cloud and valid pose; update local map. Rejected clouds never enter Room Map.

The adaptation reuses upstream IKFoM state/process/filter components and ikd-Tree. VISTA owns synchronization, initialization, deskew, correspondence construction and gates. LiDAR-to-IMU extrinsics stay fixed; online extrinsic estimation is disabled.

Two maps, two purposes

The local LIO map is a bounded registration reference, not the persistent Room Map. Keeping it small helps localization keep up; it does not discard the full room geometry saved by the mapper.

Algorithm and implementation background: FAST-LIO2, ikd-Tree and IKFoM [R01, R02, R04]. No upstream FPS claim is transferred to this hardware.

16 / Localization states and safe recovery

State and Pose Valid answer different questions; always inspect Reason.

State	Meaning / next action
WAITING IMU	No usable synchronized IMU/scan batch yet. Check source, receipt/aligned times, brackets and gaps.
INITIALIZING	Collect stable startup data and seed geometry. Keep stationary; read sample/duration/stability diagnostics.
TRACKING	LiDAR-corrected accepted pose. Pose Valid=1 allows the world mapping branch.
DEGRADED	Temporary rejection/stale correction. Live sensor view continues; no rejected world geometry is added.
LOST	Unsafe time coverage or sustained failed corrections. Hold map and last trusted pose; no IMU-only or fixed-pose substitution.

Quality rejection persists as DEGRADED for up to two acquisition seconds, then LOST. Certain timing/source faults can enter LOST immediately. After five continuous host steady-clock seconds in LOST, the worker clears stale synchronization input and retries stationary initialization.

Recovery preserves the last accepted world pose and accepted local map. With no accepted pose, an untrusted seed can be rebuilt at the configured initial pose. New geometry must still match the retained map and pass gates before publication.

Recovery boundaries

This is a local retry, not global relocalization. Moving far during LOST can prevent recovery. A latched invalid source clock may require repairing/reconnecting the driver; restarting initialization cannot invent missing timestamps.

Non-tracking telemetry shows the last trusted or initial configured pose with Pose Valid=0. It must not be interpreted as the device's current measured location.

17 / Stationary robustness and overload

Remaining still does not mean identical points arrive in every Unitree scan.

The repaired stationary check compares non-repeating samples to surfaces in a bounded sensor-frame reference, together with stable measured IMU. It no longer requires most individual XYZ points to recur near the same locations in a sparse single scan.

Safety condition	Current implementation
Startup scaling	9.81 / norm(mean stationary acceleration), fixed after initialization; published IMU and RAW remain unchanged.
Stationary support	At least 0.5 s, six-DOF-observable surfaces, 100 supported probes, 30% support, median residual below 2.5 cm and small identity-pose correction.
Bounded reference	At most 20,000 reference points and 1,000 probes per scan; mature reference frozen rather than following motion.
Velocity observation	Finite-noise zero-velocity update; not a hard fixed-pose lock and not a bypass of LiDAR quality checks.
Synchronization bounds	4,096 IMUs, four pending scans, 0.5 s overlap history; at most one expensive solve per ingestion iteration.
Overload policy	Use newest queued scan, account for skipped scans; preserve continuous IMU over skipped intervals. Prediction backlog above 2 s is unsafe.

Do not confuse stability with proof of accuracy

Ambiguous geometry and very slow/constant-speed motion can defeat stationary classifiers. Synthetic movement tests pass, but a real motion trial is still required. Do not loosen quality gates merely to replace LOST with TRACKING.

Use Release for deployment. Only the private numerical target is optimized in Windows Debug; other workers retain normal Debug behavior. Breakpoints and debugger overhead can still cause drops or missing coverage.

18 / Ground calibration and removal

Keep the floor in the Room Map; remove it only in the separate perception-oriented branch.

GroundMode	Strategy	Trade-off
static	Use the configured world floor/mount prior.	Predictable, but accuracy depends on measured mounting.
ransac	Fit a plane from startup observations and enforce floor/normal quality conditions.	Data-driven; insufficient floor evidence can leave it invalid.
hybrid	Try data-driven calibration with floor/mount priors; use static fallback when fitting fails.	Robust availability, but fallback is not a measured calibration.
none	Disable ground removal; keep the mount/world-coordinate pipeline.	Useful baseline for geometry comparison.

UseImuForGround(lidar) selects optional IMU orientation assistance. If fresh usable IMU is unavailable, ground logic uses configured mounting. This is separate from LioEnabled and does not estimate translation.

Calibration occurs at startup over configured frames. Runtime automatic ground recalibration is not a public setting. In LIO mode ground processing receives accepted world scans, so it may remain CALIBRATING while localization rejects input.

Floor visibility

When the sensor sits on a table, a limited view of the floor can make STATIC FALLBACK expected. The table surface is not automatically the true floor. Placement alone neither proves failure nor validates the fallback plane.

Robust plane-fitting foundation: RANSAC [R05]. VISTA adds its own floor priors and acceptance rules.

19 / How to judge ground quality

A green value or zero residual is not enough to certify the plane.

State	Interpretation
CALIBRATING	Collecting startup evidence; result not yet established.
VALID	A usable ground model passed the implementation checks. With static mode, validity can be configuration-based rather than a fitted proof.
STATIC FALLBACK	Hybrid could not establish an acceptable fitted plane and uses the configured static floor.
INVALID	No usable model; check mode, geometry and calibration settings.

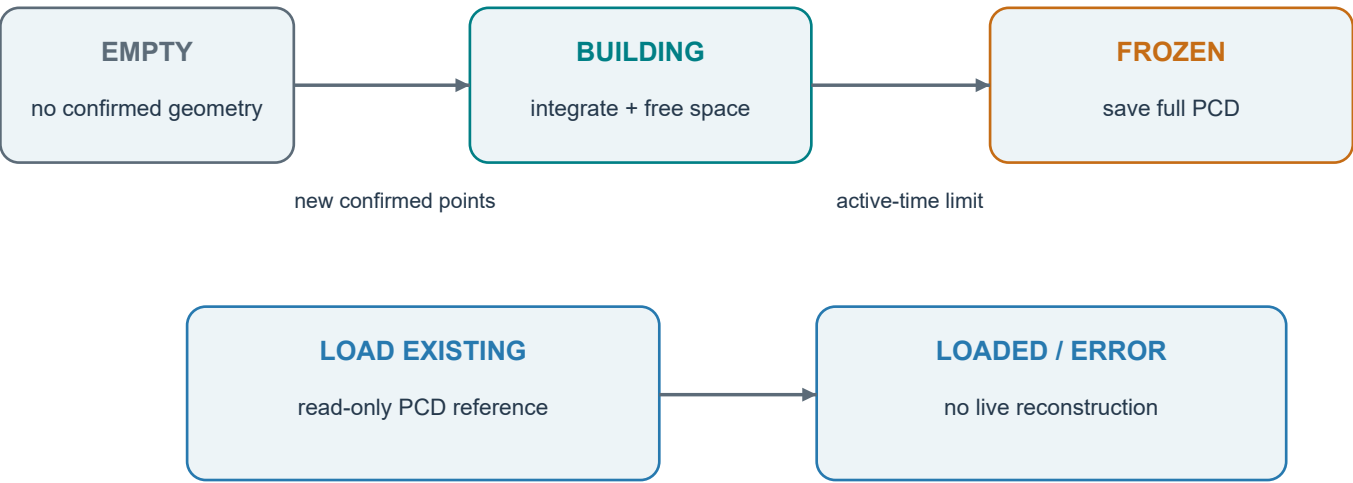
Diagnostic	Practical interpretation
Ground Tilt	Check consistency with the intended world-up direction; low tilt alone does not establish floor height.
Height Error	Difference from the expected floor/mount relationship; compare with an actual height measurement.
Ground Inliers	Fraction of supporting calibration samples. Enough support must come from the floor, not a table/wall.
RMS Residual	Plane fit spread. Compare with measurement noise and the configured threshold. Zero can mean no fit evidence.
Removed as Ground	Removed fraction for current world clouds. Excessive removal can erase useful geometry; zero may indicate no visible floor.

Recommended validation trial

Measure mount pose and optical height; capture an empty scene with visible floor/walls; compare none versus hybrid; inspect a ground plane against real geometry; record inliers/residual/height error; confirm walls/furniture remain. Repeat after changing position/orientation and restarting.

20 / Room Map lifecycle

The mapper accumulates full geometry, independently of the local odometry map.



State	Meaning
EMPTY / 0	No confirmed points yet. No-data startup does not invent a valid map.
BUILDING / 1	Integrate world hits and repeated free-space evidence.
FROZEN / 2	Stop additions and clearing; export the session PCD on transition.
LOADED / 3	Serve a valid PCD as a read-only reference; no live rebuilding.
DISABLED / 4	RoomMapEnabled=0; status/empty views may still be served.
LOAD ERROR / 5	Missing, invalid or corrupt RoomMapFile; do not silently build or overwrite it.
STORAGE ERROR / 6	Tile/checkpoint/load/export failure; inspect disk and console.

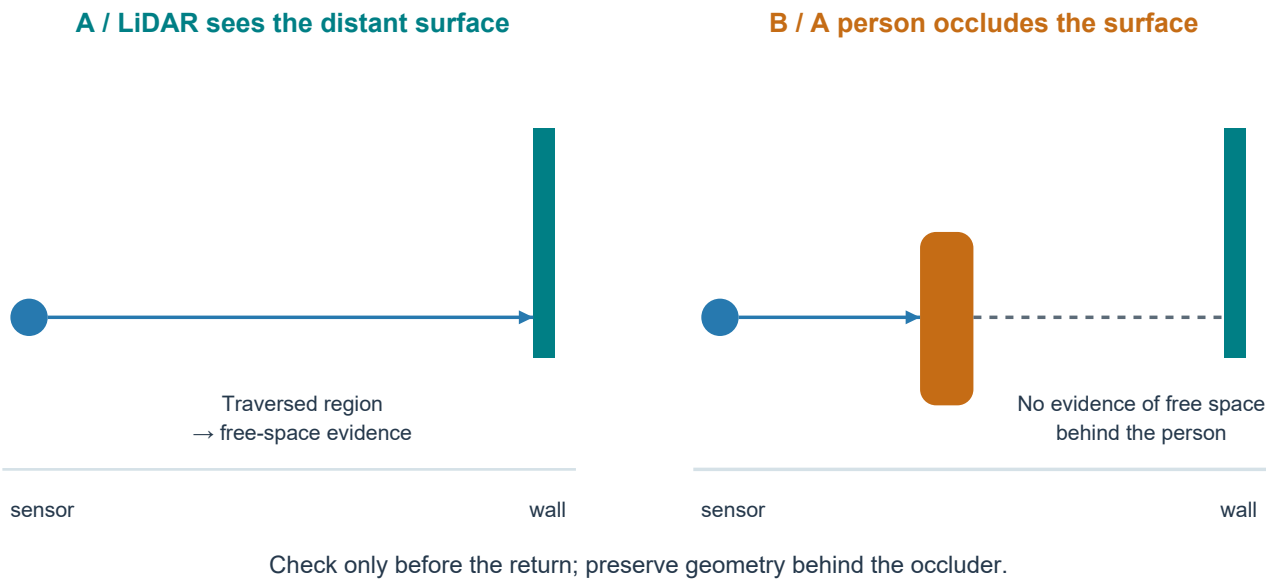
FreezeAfterSeconds counts active acquisition time. A value of 0 means continuous updating, not immediate freeze. In LIO mode invalid poses pause useful integration while preserving existing map geometry.

Loaded-map limitation

Loading a reference is not enabling detection/tracking and not relocating the sensor into the PCD. Future algorithms must explicitly establish that common frame.

21 / Hit confirmation and free-space evidence

Repeated observations suppress transient geometry; only measured free space can clear old points.



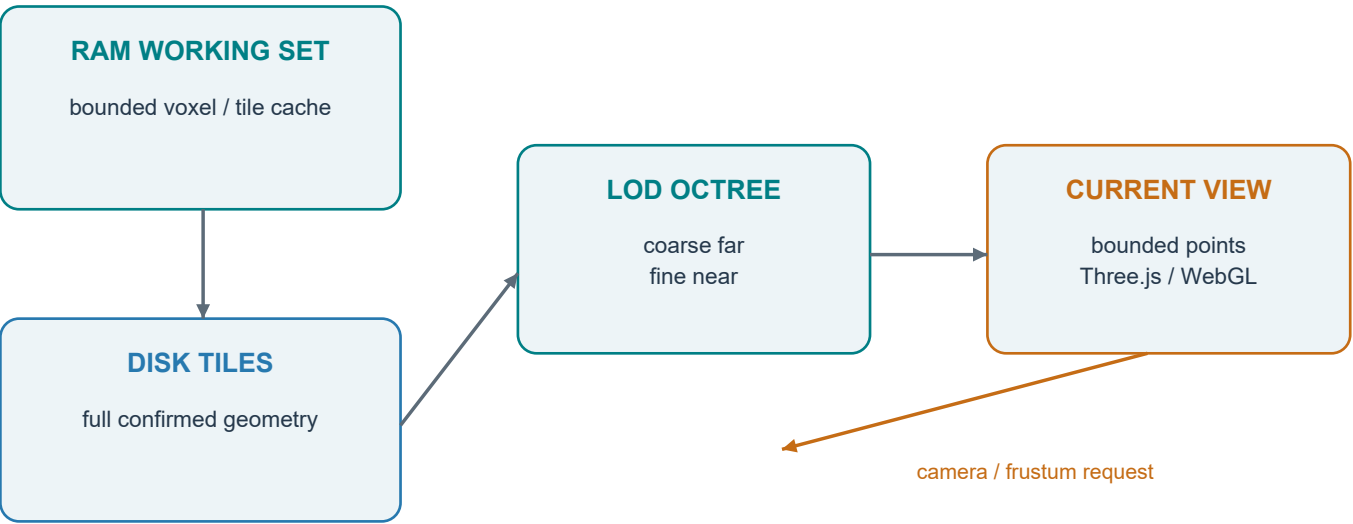
A 5 cm voxel becomes visible after three distinct hit windows by default. Many packets in one 100 ms window count once. Rays traverse cells from the acquisition-time world origin toward real returns, stopping before the measured surface.

Situation	Current behavior
New person occludes a confirmed wall	Keep points behind the person: the beam ends at the person, so hidden space is not proven empty.
Previously mapped person leaves	Later rays through the former location can clear that ghost after repeated free windows.
Person stays during acquisition	The person can still become confirmed geometry. No semantic/person detector is active.
Missing returns or missing frames	Never treat absence alone as evidence of empty space.
Same-window hit plus free evidence	The hit takes precedence; fresh hits reset free evidence.
Frozen or loaded map	No continued clearing or accumulation.

DDA traversal background [R06]. VISTA-specific confirmation windows, hit precedence and surface protection are additional rules.

22 / Large maps: RAM, disk and rendering

There is no fixed total confirmed-point cap, but every physical resource remains finite.



PCD export includes all confirmed points, not just the current view.

Budget	What it actually limits
RoomMapCacheMaxVoxels	Resident voxel records in the C++ tile cache, not the entire saved map or whole-process RAM in bytes.
RoomMapCacheMaxTiles	Resident spatial tiles, including sparse tiles; dirty pages are written before LRU eviction.
LioLocalMapMaxPoints	Local registration reference only; separate from the room map and its PCD export.
RoomMapWebSocketMaxPoints	One camera-dependent transmitted view; not the total stored geometry.
Plugin maxPoints	Browser rendering budget; effective detail is also constrained by the server budget.

VISTA uses sparse spatial tiles and octree representative samples inspired by Potree [R07]. It does not use Potree's renderer or standard files. Zoom/pan sends a camera/frustum request; the server selects a non-overlapping LOD view and the plugin replaces that view.

Scalability is not infinity

A large map can be stored without loading or drawing all points simultaneously. GPU memory, network, disk capacity and coordinate extent are still finite.

23 / Files, checkpoints and PCD export

Tile directories during BUILDING are expected; they are not a missing-map symptom.

Artifact	Purpose / completion
data/raw/YYYYMMDD_HHMMSS.bin	Optional raw payload-byte capture. Not a framed host-timing archive.
data/processed/YYYYMMDD_HHMMSS.pcd	Optional ground-processed scan log; not a deduplicated Room Map.
data/maps/RoomMap_YYYYMMDD_HHMMSS.tiles.<id>	Spatial evidence and LOD pages written while building; retained after shutdown.
data/maps/RoomMap_YYYYMMDD_HHMMSS.pcd	All currently confirmed geometry, streamed from disk tiles at freeze or clean shutdown.
Existing RoomMapFile	Loaded read-only PCD preserved; a separate view index may be created.

RoomMapPublishIntervalMilliseconds checkpoints changed pages before publishing a revision. It does not rewrite the whole PCD every second. Port 8766 sends a new revision without a second publication timer; camera/reconnect requests can still retrieve a view.

Every non-loaded building session with confirmed geometry is saved automatically. There is no RoomMapSaveEnabled switch. Freeze and shutdown use the same session file; timestamp-based names plus collision suffixes avoid overwriting another session.

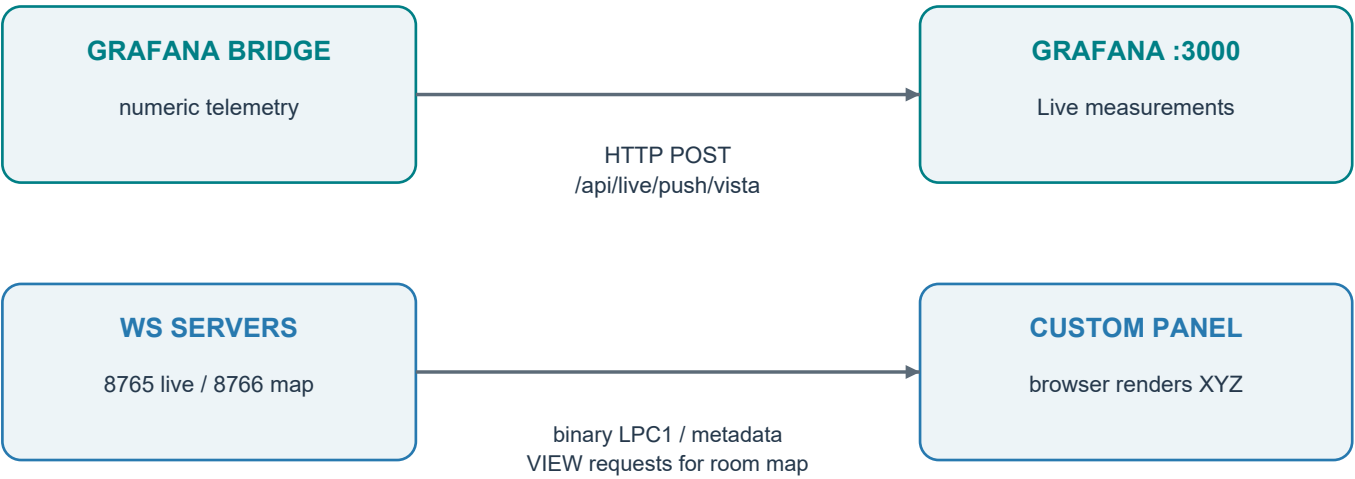
Save safety

Wait for the console's map-saved confirmation before ending the process. Stop Debugging, a kill or power loss may bypass final PCD export. Automatic recovery directly from old .tiles directories is not implemented.

PCD format reference [R08]. Stored map representatives are not every raw laser return ever observed; cleared/candidate points are not included in the confirmed export.

24 / Grafana: metrics versus geometry

The Grafana server does not render XYZ or receive the full map through the metrics bridge.



Grafana does not need the full XYZ cloud in its HTTP metrics stream.

Route	Producer / consumer
HTTP Live push	C++ bridge sends numeric/status Influx line protocol to http://host:3000/api/live/push/vista. Grafana fans out Live frames.
Live WS :8765	C++ host sends sensor-frame binary LPC1 XYZ/XYZI; the browser panel renders with Three.js/WebGL.
Room Map WS :8766	C++ host serves camera-dependent LOD; browser sends VIEW requests and renders the returned points.

lidarpointcloud is a panel plugin (ID vista-lidarpointcloud-panel, version 1.1.0). VISTA Live is a data source (ID vista-live-datasource, version 1.0.1). The dashboard UID is vistolive; the metrics namespace is vista. These four identifiers serve different purposes.

LPC1 is VISTA's little-endian point-frame payload, not a Grafana standard. Point values are binary float32; optional metadata extends the header. A connected socket can legitimately carry an empty map state.

Grafana Live/API [R09], plugin signatures [R10] and WebSocket transport [R13].

25 / Installing Grafana plugins and the dashboard

Grafana and both plugins can stay entirely on Windows.

Use Grafana 13.2.2 or later as required by the VISTA Live manifest; compatibility with later versions still needs testing. Use Grafana's actual configured plugins path; the following path is the usual local installation.

Step	Action
1. Stop and back up	Stop Grafana for replacement; back up existing plugin folders outside the scanned plugins directory.
2. Install the panel	Copy dashboard/plugins/lidarpointcloud/dist CONTENTS to C:/Program Files/GrafanaLabs/grafana/data/plugins/vista-lidarpointcloud-panel.
3. Install the data source	Copy dashboard/plugins/vistalive/dist CONTENTS to the same plugins root under vista-live-datasource (or the existing vistalive folder). No duplicate plugin IDs.
4. Allow the IDs	Edit active custom.ini as administrator; use the [plugins] setting shown below. Keep other approved IDs.
5. Restart and refresh	Start Grafana, hard-refresh the browser, verify panel 1.1.0 and data source 1.0.1 are listed.
6. Create a data source	Connections > Data sources > Add data source > VISTA Live > Save & test.
7. Import the latest JSON	Import dashboard/VISTA_Live_Dashboard_V4_Free_Space.json, select VISTA Live when prompted, keep UID vistalive and overwrite the existing dashboard.
8. Set endpoints	Live panel ws://edge-host:8765; Room Map ws://edge-host:8766. Run C++ and confirm both telemetry and geometry.

```
[plugins]
allow_loading_unsigned_plugins = vista-lidarpointcloud-panel,vista-live-datasource
```

Authorize the bridge: create a Grafana service-account token with the required push permissions [R19]. On the C++ host, keep a private GrafanaSecret.txt in the runtime working directory containing **GrafanaToken: YOUR_PRIVATE_TOKEN**, or provide VISTA_GRAFANA_TOKEN in the process environment. The local file takes precedence. Never include the real token in the dashboard, plugin, Git or team guide.

Folder name is not plugin ID. plugin.json and module.js must be directly in the installed plugin folder, not in a nested dist directory. Official signature guidance [R10].

26 / Dashboard interpretation and disconnects

No data, zero, offline and invalid pose have different meanings.

Section / panel	Behavior
Live 3D and Room Map 3D	Use direct edge WebSockets. Retain the last geometry during disconnection; a retained map is not proof of current acquisition.
Localization / Ground / Room Map	States and history tables expose quality and map lifecycle, not only connection status.
LiDAR and IMU	Online and rate/count plots describe the live measurement branch. Localization validity is separate.
Worker Health	Latest snapshot only: one row per worker, STT, Worker, Failed, Priority, Running and Seconds. Not one row per timestamp.
Ground and Room Map tables	New samples appear at the top; Room Map build/cache/I/O and bridge history use the 60,000 ms Live buffer.
System Health	Running/failed workers plus host resources and pipeline history.
Future Sensors and Applications	Reserved placeholders; no active radar/detection/tracking capability is implied.

VISTA Live expires numeric/status/table/graph frames after six seconds without a fresh bridge heartbeat. A 250 ms browser timer does this even when dashboard auto-refresh is off. It re-subscribes after errors and normal stream completion and uses ordinary frames to recover from empty results.

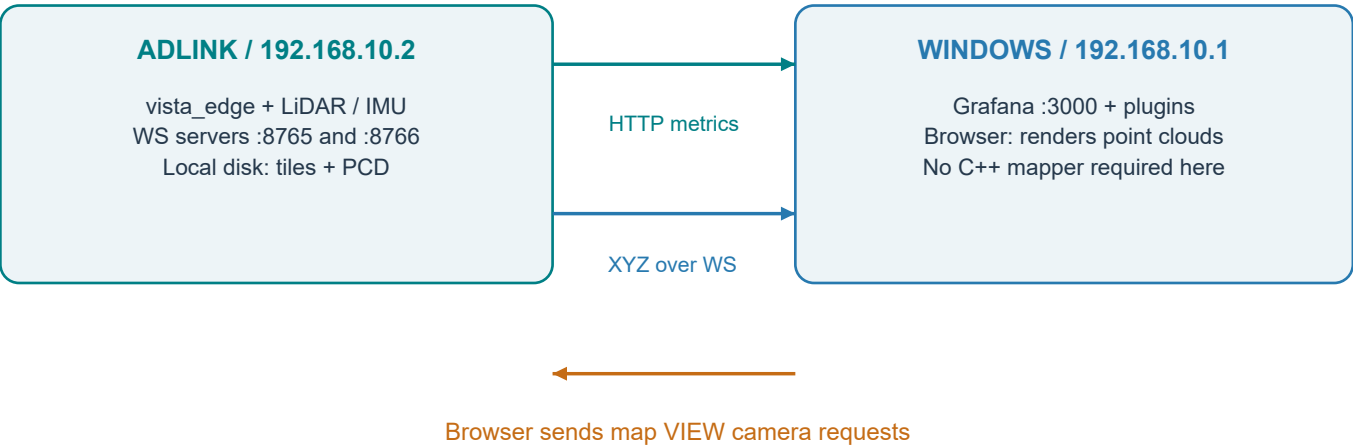
What the heartbeat establishes

This tests producer-to-Grafana liveness, not LiDAR liveness while vista_edge continues publishing. A green panel/header connection indicator is not the LiDAR Online value. Host/browser calendar clocks must agree; use a 1-2 s metrics cadence.

Live buffers are not persistent storage. Old rows age out after about 60 s; reload/server restart can discard history. A time-range selector cannot reconstruct a database that was never configured.

27 / Ethernet deployment: ADLINK to Windows

Run processing on ADLINK and Grafana/browser rendering on the Windows PC.



Ethernet carries traffic in both directions; 127.0.0.1 means this host only.

Setting / location	Value for this LAN
ADLINK DeviceConfig: GrafanaHost	192.168.10.1; GrafanaPort 3000; GrafanaNamespace vista.
ADLINK WS bind addresses	192.168.10.2 for both listeners, or 0.0.0.0 only with appropriate firewall restrictions.
Windows live panel URL	ws://192.168.10.2:8765
Windows Room Map URL	ws://192.168.10.2:8766
Windows firewall	Allow trusted ADLINK access to TCP 3000. Grafana must listen on a reachable interface.
ADLINK firewall	Allow trusted browser/Windows access to TCP 8765 and 8766.

Avoid endpoint confusion

Bind addresses belong to the server's host. Do not bind an ADLINK socket to Windows 192.168.10.1. There is no Windows WebSocket relay in the current code; relocating these servers would require another service.

Keep the LiDAR network and PC link addresses distinct if using UDP LiDAR acquisition. For HTTPS dashboards use a secured WSS/proxy setup; browsers can block insecure ws:// as mixed content.

28 / First-run validation checklist

Prove each stage independently before attempting a complete moving room scan.

Order	Check / expected evidence
1. Build identity	Confirm source revision, build type, target CPU and the runtime DeviceConfig path. Stop an old Debug process before linking the executable.
2. Acquisition	Valid decoded counts, Unitree IMU rate and no competing COM owner. Compare input and live filtered counts.
3. Stationary baseline	LioEnabled=0; keep the device fixed; live cloud works and map confirms visible floor/walls.
4. Ground baseline	Measure mounting; compare none/static/hybrid; inspect diagnostics rather than relying on color.
5. LIO startup	Enable LIO, verify extrinsics/time metadata; hold stationary through initialization. Tracking and Pose Valid=1 are required for world map updates.
6. Slow movement	Record a short translation/rotation with clear geometry. Compare pose against measured physical movement and look for doubled walls.
7. Map lifecycle	Test continuous building, finite active-time freeze, map-saved confirmation and PCD content.
8. Read-only reload	Load the exact saved PCD; expect LOADED and no overwrite or rebuild.
9. Reconnect test	Stop/restart vista_edge and interrupt network: metrics clear after six seconds, resume; 3D retains geometry and reconnects.
10. Target trial	Repeat on ADLINK Release, observe CPU/RAM/disk/drops, then conduct endurance testing.

Reproducible experiments

Before comparing old/new versions, use the same capture, coordinate frame, filters, scan duration and render budget. A screenshot alone cannot locate the stage where points were lost.

29 / Configuration: device and ground

Values below are the inspected source TXT, not universal sensor calibration.

Keys	Snapshot / meaning
Lidar	unitree-l2; one selected LiDAR type per run.
RawLoggingEnabled(Lidar)	1; append raw capture. PointCloudLoggingEnabled(Lidar)=0 disables processed scan logging.
ReconnectIntervalSeconds(Lidar)	5; device reconnection delay, not browser retry.
ConnectionMode / SerialPort / BaudRate	auto / COM6 / 4000000. Set serial mode and the actual COM/Linux path for a deliberate serial-only deployment.
SensorIP / SensorPort	192.168.1.3 / 4141 in source TXT; verify the actual UDP/TCP device settings before network use.
LocalIP / LocalPort(unitree-l2)	192.168.1.2 / 6201; these must match a local LiDAR interface, not the Grafana host.
GroundMode / UseImuForGround(lidar)	hybrid / 1; fit plus fallback, with optional IMU orientation assistance.
MountX/Y/Z(lidar)	0 / 0 / 0.7 m; measure the actual optical-center pose.
MountRoll/Pitch/YawDeg(lidar)	0 / 0 / 0 degrees; rotation convention must match sensor mounting.
FloorZ(lidar)	0 m in world coordinates.
GroundDistanceThreshold(lidar)	0.10 m: plane-distance classification/fit tolerance.
GroundNormalToleranceDeg(lidar)	15 degrees: accepted normal departure from expected up.
GroundCalibrationFrames(lidar)	50 startup world frames.
GroundMinInlierRatio(lidar)	0.10 in current TXT; support threshold, not a guaranteed floor classification.

Parenthesized labels are part of the supplied configuration syntax. Restart after edits. Runtime preprocessing range/ROI/voxel settings are described in section 13. L515-specific TXT settings select USB serial, depth width/height and frame rate; they are inactive with Unitree selected.

30 / Configuration: LIO geometry and work budgets

Local registration detail and permanent map detail need not use the same voxel size.

Key	Snapshot	Effect
LioEnabled	1	Moving-pose branch on. 0 uses fixed-mount world geometry.
LioLidarToImuTranslationX/Y/Z	0.007698 / 0.014655 / -0.00667 m	LiDAR pose IN IMU frame; example SDK static TF, not a verified calibration for every L2/external IMU.
LioLidarToImuRoll/Pitch/YawDeg	0 / 0 / 0	Fixed rotation extrinsics; degrees.
LioScanVoxelSizeMeters	0.05 m	Downsample current matching scan. Larger reduces work/detail.
LioMapVoxelSizeMeters	0.20 m	Density of bounded local registration map. Not persistent Room Map resolution.
LioLocalMapRadiusMeters	30 m	Neighborhood retained for odometry; not room storage extent.
LioLocalMapMaxPoints	50000	Local map point budget. Larger costs memory/search time.
LioScanMaxPoints	5000	Maximum matching scan points; not all live/Room Map observations.

Why not merge the voxel settings?

Scan voxels select observations used in a solve. Local-map voxels control search reference density. RoomMapVoxelSizeMeters controls persistent reconstruction detail. Sharing one knob would couple CPU load, registration quality and final map detail unnecessarily; changing one task could degrade another.

Snapshot note

The historical scan example was 0.15 m. The current file uses 0.05 m; guide readers should not copy a historical chat value without comparing the runtime configuration.

31 / Configuration: LIO initialization and gates

Relaxing a threshold can accept a wrong pose; diagnose the reason first.

Key	Snapshot	Meaning
LioInitializationSamples	50	Valid stationary IMU samples; minimum duration/stability checks still apply.
LioMaxImuGapMilliseconds	100	Largest permitted IMU gap in covered intervals; no infinite extrapolation.
LioSynchronizationTimeoutMilliseconds	500	Wait for required coverage before reporting missing/gapped input.
LioMinMatchRatio	0.15	Required effective correspondence fraction. Not scene accuracy.
LioMaxResidualMeters	0.15 m	Maximum accepted match residual. Larger tolerates noise and also bad fits.
LioMaxTranslationStepMeters	2 m	Maximum accepted pose translation step; not a commanded movement allowance.
LioMaxRotationStepDeg	30 degrees	Maximum accepted rotation step.

Geometry observability, finite-state and covariance checks also exist in code. They cannot be replaced by changing match ratio alone. A flat floor can constrain height/tilt but leave other motion directions weakly observable.

Use a diagnostic-first adjustment order

- Confirm actual measurement timing, acceleration including gravity, gyro units and fixed extrinsics.
- Use Release and measure processing/drops before increasing scan or map budgets.
- Improve scene geometry and mounting; inspect detailed Reason values.
- Change one setting at a time on recorded stationary AND moving trials, checking drift and wall consistency.

Recovery policy

Five seconds of continuous LOST triggers a code-level recovery policy, not a new DeviceConfig key. It is not proof that localization has recovered.

32 / Configuration: Room Map geometry and lifecycle

Mapping evidence is counted in observation windows, not simply in packets.

Key	Snapshot	Meaning
RoomMapEnabled	1	Create/use map; disabled status can still be published.
RoomMapVoxelSizeMeters	0.05 m	Persistent cell size. Smaller increases detail, evidence storage and work.
RoomMapMinObservations	3	Distinct hit windows before confirmation.
RoomMapObservationIntervalMilliseconds	100	Monotonic window length for hit/free evidence.
RoomMapFreezeAfterSeconds	0	Continuous. Positive values freeze after active acquisition seconds.
RoomMapPublishIntervalMilliseconds	1000	Changed-map checkpoint/revision and bounded repeated view-request cadence; not full-PCD export cadence.
RoomMapLoadExisting	0	Build new. Set 1 to serve a read-only existing PCD.
RoomMapFile	blank	Valid in build mode; load mode requires the exact existing PCD path.

Build mode ignores RoomMapFile as an output name and allocates a timestamped session under data/maps. A loaded map does not subscribe to cleaned geometry for rebuilding. Changing mounting or LIO origin does not automatically register live data to that reference.

Removed settings

RoomMapSaveEnabled, RoomMapControlFile, RoomMapMaxVoxels and AutoGroundRecalibration are obsolete for this workflow. Do not restore old keys from earlier examples.

33 / Configuration: cache, LOD and free space

Balance spatial locality, rendering detail and ray work instead of raising every budget.

Key	Snapshot	Meaning
RoomMapCacheMaxVoxels	250000	Resident voxel-record budget, not whole-map cap.
RoomMapCacheMaxTiles	128	Resident tile count budget.
RoomMapTileSizeMeters	2 m	Spatial tile edge, rounded to whole voxel counts.
RoomMapLodPointsPerNode	128	Octree representative samples per node.
RoomMapFreeSpaceMinObservations	5	Distinct free windows before clearing a confirmed cell; conservative.
RoomMapRaycastMaxRaysPerWindow	256	Sampled rays per window; not every beam is traversed.
RoomMapRaycastMaxRangeMeters	30 m	Maximum free-space traversal range, not hit fusion range.
RoomMapRaycastSurfaceMarginMeters	0.10 m	Protect measured surfaces; implementation preserves at least one voxel diagonal.

A tile must fit the cache budget and contain at most 64 cells per axis. At 2 m / 0.05 m, a dense tile is $40^3 = 64,000$ cells. More sparse tiles can still use I/O/cache capacity. The LOD reader has a separate bounded 512-node cache.

Ray traversal also has a hard 250,000-step window limit. Counters report skipped work rather than allocating unknown/free space indefinitely. More rays can accelerate ghost cleanup but may increase CPU, locking and disk work.

Evidence depends on visibility

Three hits in 100 ms windows do not guarantee confirmation at exactly 300 ms; the sensor must revisit that voxel in different windows. Five free windows do not guarantee clearing at 500 ms either.

34 / Configuration: telemetry and WebSockets

Three publication intervals regulate three different outputs.

Configuration group	Snapshot / meaning
GrafanaEnabled	true gates Grafana Live AND both WebSocket workers. No separate public WS enabled switches.
GrafanaHost / Port / Namespace	127.0.0.1 / 3000 / vista for local Windows; use Windows IP for ADLINK deployment.
GrafanaPublishIntervalMilliseconds	2000 ms: numeric/status HTTP batch cadence.
GrafanaRetryIntervalSeconds	5 s: bridge retry; not sensor or browser retry.
SystemMonitorIntervalMilliseconds	2000 ms: host/worker monitoring cadence.
PointCloudWebSocketBindAddress / Port	127.0.0.1 / 8765 in local source TXT.
PointCloudWebSocketMaxPoints / MaxClients	100000 points per frame / 4 concurrent clients on that listener.
PointCloudWebSocketPublishIntervalMilliseconds	100 ms: live frame throttle, up to roughly 10 sends/s when input allows.
PointCloudWebSocketRetryIntervalSeconds	2 s: server retry after listener errors.
RoomMapWebSocketBindAddress / Port	127.0.0.1 / 8766 in local source TXT.
RoomMapWebSocketMaxPoints / MaxClients	100000 points per camera view / 4 concurrent clients.
RoomMapWebSocketRetryIntervalSeconds	2 s: map-listener retry; browser reconnection options are separate.

RoomMapPublishIntervalMilliseconds=1000 regulates changed-map checkpoints and view revisions. It is neither the metrics interval nor a timer to append raw scans to the browser. MaxClients counts browser/panel connections, not the number of WebSocket servers you may implement.

Keep Grafana publish cadence below the six-second expiry threshold. Tokens belong in local secret storage/environment, not DeviceConfig, JSON, plugins or this document.

35 / Building on Windows

The executable path depends on the selected generator, build directory and configuration.

Prerequisites: Git with recursive submodules, Visual Studio C++ toolchain and compatible CMake. The supplied preset selects Visual Studio 2019 x64; open `embedded_C` as the CMake project in Visual Studio and select `vista_edge` as the startup target.

```
git submodule update --init --recursive
cd embedded_C
cmake --preset windows-vs2019
cmake --build --preset windows-debug
ctest --preset windows-debug --output-on-failure
```

Item	Path / action
Debug executable	<code>embedded_C/out/build/windows-vs2019/Debug/vista_edge.exe</code>
Debug unit tests	<code>embedded_C/out/build/windows-vs2019/Debug/vista_edge_tests.exe</code>
Release	Use the same configured multi-config build with <code>--config Release</code> ; executable is in its Release subfolder.
Runtime files	<code>DeviceConfig.txt</code> is read from working directory. Confirm the startup config summary and Visual Studio launch settings.
LNK1168	An existing process may hold <code>vista_edge.exe</code> open. Stop your Debug session, then build; do not assume tests relinked the app.

The FAST-LIO numerical target uses `/O2 /Ob2` with `/Zi` in Debug and omits `/RTC1` only there. Other VISTA sources retain normal Debug checks. Optimized numerical variables may be unavailable or move while debugging.

Latest small cleanup

The old "Multi thread started" console line is removed from the generated `ikd-Tree` adaptation. Tree initialization still works; no upstream checkout or thread policy is changed by suppressing that log.

36 / Modern Linux and dependency isolation

Compile for the target OS/CPU; a Windows executable cannot run on Linux.

```
git submodule update --init --recursive
cd embedded_C
cmake --preset linux-x86_64-gcc
cmake --build --preset linux-x86_64-debug
ctest --preset linux-x86_64-debug --output-on-failure
```

For a modern native ARM64 system use linux-arm64-gcc and linux-arm64-debug. These presets require CMake 3.20 or newer; they are not the Ubuntu 18.04/CMake 3.10 path. Linux binaries are named vista_edge and vista_edge_tests, without .exe.

Dependency	Integration
FAST-LIO / ikd-Tree / IKFoM	Pinned source; CMake generates portable private adaptations and builds vista_fast_lio. No ROS/PCL runtime.
Eigen 3.4 / Boost.Preprocessor	Pinned headers for the numerical implementation; not a dependency on the full Boost runtime.
librealsense 2.50.0	Optional static SDK build. ARM preset selects RSUSB by default; legacy camera-free build disables it.
Portable filesystem	Pinned ghc/filesystem v1.5.16 used when standard C++ filesystem header/link checks fail; consistent backend across VISTA targets.

Generated copies replace a small set of upstream POSIX/PCL/OpenMP-specific glue without editing submodules. The hidden asynchronous tree rebuild is disabled; the owning LIO worker controls updates.

Licensing

Derived FAST-LIO code carries upstream GPLv2 obligations; preserve notices and review distribution requirements. Editor choice, ROS-free operation and static linking do not remove license requirements.

Traditional CMake and submodule mechanics [R11-R12]; source licenses are in third_party and the adapted core.

37 / ADLINK Ubuntu 18.04 deployment

Use the legacy native script; do not upgrade the Jetson software stack solely to satisfy presets.

Reported device environment	Value
Device / CPU architecture	ADLINK NEON-2000-JNX / aarch64
OS / BSP	Ubuntu 18.04.5 LTS / L4T R32.5.0 reported by nv_tegra_release
Compiler / CMake	g++ 7.5.0 / CMake 3.10.2
Other tools	Git 2.17.1, Python 3.6.9, CUDA 10.2, OpenCV 4.1.1
Memory	7.6 GiB RAM, 3.8 GiB swap in the reported snapshot.

```
# From the cloned repository root on ADLINK
git submodule update --init --recursive
bash embedded_C/scripts/build_adlink.sh
cd embedded_C/out/build/linux-adlink-bionic
# Edit this directory's DeviceConfig.txt, then run:
./vista_edge
```

The script uses Release, tests ON, RealSense OFF and two build jobs. VISTA_BUILD_JOBS=1 reduces parallel build memory. It uses traditional CMake 3.10 commands and does not require Python, CUDA, ROS, presets or changing the system compiler.

Configure the actual Linux serial path and permissions, Windows Grafana host, LAN bind addresses and token. Run from the build directory so runtime config is found. Back up deployment config: rebuilding can replace copied files.

Hardware validation remains necessary

Native compilation or an Ubuntu 18.04 ARM64 sysroot avoids assuming newer Linux binaries are ABI-compatible. QEMU tests do not validate real serial devices. Preserve the vendor-supported BSP; assess updates with ADLINK/NVIDIA guidance [R14].

38 / Verification evidence and limitations

Results below are recorded implementation checks, not new hardware benchmarks performed for this PDF.

Complete suite: 191 C++ tests	Execution environment
Windows x64 / VS2019 Debug	Native Windows.
Linux x86-64 / GCC 13 Debug	Ubuntu 24.04 WSL.
Legacy x86-64 / GCC 7.5 / CMake 3.10	Ubuntu 18.04 userspace in WSL; optional L515 off.
Legacy ARM64 / GCC 7.5 cross-build	Ubuntu 18.04 ARM64 userspace with QEMU; optional L515 off.

The recorded suite covers synchronization, clocks, independent models, deskew, stationary/moving synthetic rooms, initialization/recovery, retained map continuity, geometry rejection, storage/LOD, protocol and telemetry contracts. The latest console-log suppression also passed the full suite on these four targets.

Stationary RAW estimator audit	Accepted / examined	LOST	Max displacement
20261006_192944.bin / 530.196 s	12563 / 12578	0	0.0235 m
20261006_192542.bin / complete	1206 / 1221	0	0.0405 m
Latest / first 60 s / every fifth scan	279 / 284	0	0.0576 m
Latest / first 60 s / every tenth scan	138 / 142	0	0.0369 m

Evidence boundaries

Displacement is relative to an offline initial anchor, not external ground truth. RAW does not retain host receipt timing: this cannot prove live USB/debug scheduling or actual topic-drop behavior. No real movement, ADLINK hardware benchmark or 24/7 soak result is claimed.

The test runner executes all internal cases by default; ctest may report fewer test executables than internal cases. A passing suite is necessary but not a substitute for calibration and trajectory validation.

39 / Troubleshooting: identify the failing stage

Check transport, timing, estimation, accumulation and rendering separately.

Symptom	Evidence to collect / action
All numeric panels become No data	Bridge heartbeat age, HTTP errors/token, VISTA Live selection/version, browser/Grafana calendar clocks, stream reconnection.
Green header but no values	Connection is not data. Inspect channel namespace, field types, heartbeat and query/transform contracts.
Live 3D disconnected	Correct browser-visible WS URL, server listener/bind, firewall, active GrafanaEnabled and client limit.
Live works, Room Map empty	Pose Valid, Room Map state, confirmed/candidate cells, cleaned world inputs and LIO Reason.
LIO remains INITIALIZING	Sample duration/count, measured accel/gyro stability, units, input timing and visible geometry.
LIO LOST while stationary	Detailed Reason: brackets/gaps, horizon, backlog, source/reset, match support/residual/covariance. Compare recorded input; do not only relax gates.
Ground stays CALIBRATING	Does it receive accepted world clouds? LIO rejection can starve this branch while live data continues.
Map looks small or builds slowly	Repeated hit-window visibility, scan pattern, occlusion, world filters, confirmed count, LOD camera/budgets, input drops and disk/cache times.
Only tile folders, no PCD yet	Expected during BUILDING; freeze or graceful stop, then wait for map-saved confirmation.
LOADED fails	Exact accessible PCD path/header; LOAD ERROR is intentional, not automatic rebuild.

40 / Performance, safety and long-term operation

Measure on the target device instead of inferring a fixed RAM estimate from the voxel count.

Concern	Existing mechanism / remaining work
CPU and latency	Bounded matching, latest-scan overload policy, ray budgets and LOD. Profile Release on ADLINK; Debug pauses are not normal acquisition.
RAM	Bounded local map, topic rings, tile/LOD caches and views. Track resident memory; limits count records, not whole-process bytes.
Disk	Out-of-core geometry and optional logs can grow indefinitely in time. Add retention/quota/alerts; RAW/processed logs do not have demonstrated rotation.
Map speed	Tile-ordered updates and evidence-only writes reduce churn. Observe integration/checkpoint/view and lock-wait timing before changing budgets.
Shutdown/power loss	Cooperative stop exports PCD. Transactional recovery from partial tile/index writes is not implemented.
Network access	Direct WS is not an authenticated public service. Restrict LAN/firewall, secure remote routes and use HTTPS/WSS where required.
Secrets and privacy	Do not commit tokens; rotate exposed credentials. Point clouds can still reveal people/activity without camera images.
Production validation	Fault injection, disconnect tests, disk-full tests, motion ground truth, graceful-stop timeout and endurance trials remain necessary.

Do not increase cache, client, frame and scan budgets together. Keep a baseline config and change one knob per replay/target trial. Additional clients multiply view/network work even if the stored map is shared.

Operational boundary

The current dashboard is a development/observability tool, not a safety-certified human detection or collision-avoidance system.

41 / Roadmap: detection, tracking and the digital twin

Keep future perception independent from the persistent full-geometry map.

Next phase	Required output / validation
1. Geometry baseline	Record calibrated stationary and moving datasets, ground truth, pose/map drift, sensor noise and timing quality.
2. Noise removal and clustering	Evaluate filters and spatial/adaptive clustering without erasing thin surfaces or sparse objects. Clusters are candidates, not semantic people.
3. Detection	A typed object message with centroid, 3D box, class/confidence and timestamp; benchmark indoor Unitree data.
4. Tracking	Timestamp-aware motion model, association, ID lifecycle, velocity/direction and uncertainty.
5. Moving-sensor association	Local detection can use sensor cloud while LIO is invalid; world tracking needs a valid pose or explicit uncertainty/hold policy.
6. Background map comparison	Loaded/frozen room geometry can help identify change, but registration and dynamic/static classification must be implemented.
7. Analytics and fusion	Occupancy, zones, dwell/flow and camera/radar fusion with independent messages and calibrated clocks/extrinsics.
8. Global consistency	Loop closure/relocalization and map versioning if large-area or repeated trajectories require consistent geometry.

The intended future replacement of a ground-removed display by moving-object points is a product direction, not a feature in today's code. Preserve the independent measurement branch so localization faults do not hide raw perception input.

Starting points: noise filtering [R15], AB3DMOT [R16], SimpleTrack [R17] and HOTA evaluation [R18]. Their benchmark results are not VISTA indoor performance claims.

42 / Glossary and common misunderstandings

Use these terms consistently in reviews and experiments.

Term	Meaning in this project
Point cloud / scan	A set of current measured XYZ points; not a solid mesh or full-room snapshot.
Pose	Position and orientation in a defined coordinate frame; validity/uncertainty matters.
IMU	Acceleration and angular velocity sensor; gravity and bias must be modeled. Quaternion is orientation, not position.
Deskew	Compensate sensor movement between individual point acquisition times.
LIO	Local LiDAR-inertial odometry; not automatically global SLAM.
Ground removal	Floor separation for object-oriented processing; not the source of the full-geometry Room Map.
Voxel / tile	A spatial evidence cell / a disk-cache grouping of cells. Not interchangeable with one original laser return.
Candidate / confirmed	Geometry awaiting repeated evidence / geometry meeting the hit-window rule.
Free space / occlusion	A ray-proven empty segment / an unobserved region behind a return. Occlusion is not free space.
LOD	View-dependent detail selection; store more than is drawn at once.
PCD	Point-cloud file format. A map export is confirmed representative geometry, not the entire RAW history.
Grafana Live / VISTA Live	Grafana streaming service / the project data source wrapper with expiry/reconnect behavior.
UID / namespace / plugin ID	Dashboard identity vistalive / stream namespace vista / plugin manifest identity.
Digital twin	Persistent geometry plus future semantics, dynamic state and history; the current project is a foundation.

43 / Repository evidence and further reading

The inspected working tree is the authority for implementation-specific behavior.

Primary repository evidence	What it supports
<code>main.cpp</code> ; <code>config.cpp/.hpp</code> ; <code>DeviceConfig.txt</code>	Worker count/priority, public keys, defaults, initial wiring and current numeric snapshot.
<code>models/topics.hpp</code> ; <code>models/README.md</code>	Independent IMU/mapping models, topic contracts and timing metadata.
<code>4_Applications/mapping/lio/README.md</code> + core	ROS-free adaptation, synchronization, initialization, stationary constraints, five-second recovery and recorded tests.
<code>4_Applications/processing/pointcloud</code>	Independent sensor/world branches, ground modes and diagnostics.
<code>4_Applications/mapping/room_map</code> ; <code>2_Transport/storage</code>	Evidence, tiles/LOD, map lifecycle/checkpoints, load/export behavior.
<code>4_Applications/grafana_bridge</code> ; <code>pointcloud_websocket</code>	Metrics formatting, measurement subscriptions and geometry servers.
<code>dashboard/VISTA_Live_Dashboard_V4_Free_Space.json</code>	UID, selected fields, layout, latest worker snapshot and time-history transformations.
<code>dashboard/plugins/*/src/plugin.json</code> + <code>INSTALL.md</code>	Panel 1.1.0 and data source 1.0.1 IDs, dependencies and installation.
<code>CMakePresets.json</code> ; <code>scripts/build_adlink.sh</code> ; <code>cmake</code>	Modern and legacy build paths, generated adaptations and filesystem fallback.
<code>unittest</code> ; recorded Oct 6 checks	191 cases across four target environments; replay results and their limits.

The following references explain foundations, official interfaces and future options. A citation does not mean VISTA imports that package or reproduces every method in the paper. Links are clickable; version-specific repository dependencies should stay pinned.

Reference access and scope

References were checked where accessible on 6 October 2026. Publisher DOI endpoints for the older RANSAC/DDA papers were not retrievable; DOI links are retained as bibliographic entry points. External documentation can change.

44 / References: localization and geometry

Research foundations and upstream implementation sources.

[R01] FAST-LIO2: Fast Direct LiDAR-inertial Odometry

W. Xu, Y. Cai, D. He, J. Lin and F. Zhang. arXiv:2107.06829, 2021.

Direct point-to-map registration with an iterated filter. VISTA is a ROS-free adaptation with its own timing/gates.

[Open source](#) / [publication](#)

[R02] ikd-Tree: An Incremental K-D Tree for Robotic Applications

Y. Cai, W. Xu and F. Zhang. arXiv:2102.10808, 2021.

Incremental search-map maintenance. VISTA disables the hidden background rebuild in generated portable copies.

[Open source](#) / [publication](#)

[R03] Unitree LiDAR L2 SDK

Unitree Robotics. Official unilidar_sdk2 source and examples.

Protocol, IMU and static-transform reference; verify calibration for the actual hardware.

[Open source](#) / [publication](#)

[R04] IKFoM toolkit and FAST-LIO source

HKU MARS. Upstream filter toolkit; FAST-LIO is pinned in third_party.

Filter/state model background and preserved source licenses; not a ROS runtime requirement.

[Open source](#) / [publication](#)

[R05] Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography

M. A. Fischler and R. C. Bolles. Communications of the ACM 24(6), 381-395, 1981.

Robust model-fitting foundation; does not prove a detected plane is the true floor.

[Open source](#) / [publication](#)

[R06] A Fast Voxel Traversal Algorithm for Ray Tracing

J. Amanatides and A. Woo. Eurographics, 1987.

DDA voxel traversal. Multi-window clearing and endpoint protection are project-specific rules.

[Open source](#) / [publication](#)

45 / References: mapping, dashboard and deployment

Use official APIs and formats alongside the project's actual contracts.

[R07] Potree: Rendering Large Point Clouds in Web Browsers

M. Schuetz. TU Wien master thesis, September 2016.

Hierarchical sampling and view-dependent LOD inspiration; VISTA uses custom tiles/octree, not Potree format.

[Open source](#) / [publication](#)

[R08] The PCD (Point Cloud Data) file format

Point Cloud Library. Official format documentation.

Header/field/ASCII format background. Citing PCD does not imply VISTA links PCL.

[Open source](#) / [publication](#)

[R09] Set up Grafana Live

Grafana Labs. Official streaming and HTTP Push API documentation.

Influx line protocol, channels and browser subscriptions; Live is not a persistent database.

[Open source](#) / [publication](#)

[R10] Plugin signatures

Grafana Labs. Official plugin-management documentation.

Allow local unsigned plugin IDs deliberately; preserve other approved IDs and review trust.

[Open source](#) / [publication](#)

[R11] cmake(1), version 3.10

Kitware. Official legacy CMake command reference.

Traditional build-directory syntax used by the native ADLINK script; presets need newer CMake.

[Open source](#) / [publication](#)

[R12] Git submodule manual

Git project. Official documentation.

Recursive initialization and pinned third-party revisions for reproducible builds.

[Open source](#) / [publication](#)

46 / References: operations and future perception

These sources guide validation and future work; they are not implemented tracker claims.

[R13] The WebSocket Protocol

I. Fette and A. Melnikov. IETF RFC 6455, December 2011.
Transport/handshake/close semantics; LPC1 is the project-specific payload layered on top.

[Open source](#) / [publication](#)

[R14] NVIDIA L4T 32.5 archive

NVIDIA. Official platform/BSP release information.
Context for the reported Ubuntu 18.04-based device stack; coordinate upgrades with ADLINK support.

[Open source](#) / [publication](#)

[R15] Removing outliers using a StatisticalOutlierRemoval filter

Point Cloud Library. Official filtering tutorial. Future reading.
Noise-removal candidate to evaluate on indoor Unitree data; not claimed as an active stage.

[Open source](#) / [publication](#)

[R16] 3D Multi-Object Tracking: A Baseline and New Evaluation Metrics

X. Weng, J. Wang, D. Held and K. Kitani. arXiv:1907.03961; AB3DMOT, IROS 2020.
Transparent tracking-by-detection baseline; detector quality and indoor domain transfer remain separate.

[Open source](#) / [publication](#)

[R17] SimpleTrack: Understanding and Rethinking 3D Multi-object Tracking

Z. Pang, Z. Li and N. Wang. arXiv:2111.09621, 2021.
Study preprocessing, association, motion models and track lifecycle separately.

[Open source](#) / [publication](#)

[R18] HOTA: A Higher Order Metric for Evaluating Multi-Object Tracking

J. Luiten et al. arXiv:2009.07736, 2020; IJCV, 2021.
Evaluate detection and association quality, not only FPS or visually plausible tracks.

[Open source](#) / [publication](#)

[R19] Service accounts

Grafana Labs. Official machine-access authentication documentation.
Create and restrict bridge credentials; separate private token storage from public configuration.

[Open source](#) / [publication](#)